



Our second battle takes place between two nVidia brethren: the GeForce2 Ti VX and the GeForce4 MX420 chips. The GeForce4 tag on the latter chipset is a misnomer. In reality the GeForce4 MX420 suffers from a great deal of ignominy due to the fact that it hardly carries any of the GeForce4 chipset features. Nevertheless, it does manage to show its teeth and beats all of the chipsets mentioned in this test. It scores a wonderful 152.8 fps.

We finally finish this round up with the GeForce2 Ti VX. This chipset packs a pretty mean punch and finishes a close second to the GeForce4 MX420 with 134.5 fps. Cards based on this chip can be considered pretty affordable and thus we can safely recommend this chipset to those of you who are looking for a cheaper alternative to the GeForce4 MX420 chipset.

Maximum quality: To put the latest graphics chipsets through their paces and evaluate their performance, a resolution of 1024x768 is a must. The high quality of visuals that this mode delivers rules out every other chipset except for the latest generation GPUs. The maximum quality test is useful to evaluate the memory bandwidth of a chipset, since the textures are now much larger than those used in the previous modes.

This is where things get interesting, as the GeForce2 MX400 leaves us and the big guns from nVidia and ATi decide to crash the party.

An interesting observation that this test offers is that almost all the cards from the mid-range pass our 60 fps hurdle to join us in this final round. Some of these chipsets are based on old technology while others are brand-spanking new, with cool features to boot and a promise of a healthy life-cycle. This is what will separate the grain from the chaff in this test, we think. Remember, we will assume that budget is a thing of the past and you are looking for the best. If you are still restricted by the green stuff, then by all means go for one of the mid-range chipsets—they should serve you just fine.

With that out of the way, we decide to shift our attention to the ATi Radeon 8500, nVidia's GeForce3 Ti 500, GeForce4 Ti 4200 and GeForce4 Ti 4600. All the chipsets that will now be discussed are known as GPUs, short for Graphical Processing Units since they are programmable. This is where the future lies and if you are spending a significant amount of money on a card based on one of these chipsets, you need to be assured that future games will not leave it panting.

Let us start with the ATi card. The Radeon 8500 is future proof, and John Carmack, the id Software programming guru, reckons that this chipset will be more than capable of running the highly-anticipated *DOOM III*. The Radeon 8500 also offers an excellent anti-aliasing and anisotropic filtering solution—features that help enhance the 3D quality of a

game. However, the fact remains that ATi has iffy drivers, as it will soon be evident. Considering the superior architecture and the price that you pay for this card, one would expect it to run alongside the GeForce4 chipsets, but we observe otherwise (it scores only 143.3 fps). Also noteworthy is that a card based on an ATi chipset will always offer excellent 2D and 3D quality. Moreover, the Radeon 8500 accelerates your DVD for crisp and clear viewing. If you find these facts alluring, then by all means go for this chipset.

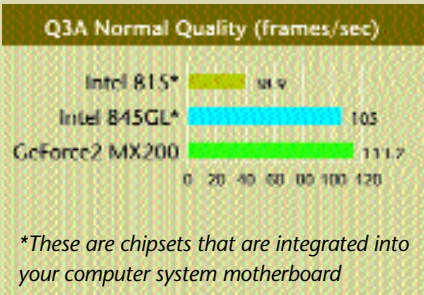


The performance of the GeForce3 Ti 500 chipset brings into question the driver quality of the ATi chips. Although inferior to the 8500, the Ti 500 is almost 6 fps ahead of the former (149.2 fps). The possibility that current games are not designed to get the most out of the brand new Radeon 8500 goes for a toss as we see the architecturally similar GeForce4 Ti 4200 and Ti 4600 do pretty well for themselves.

Too well in fact, they burn right through everything. Posting an amazing 231.8 fps the GeForce4 Ti 4200 puts everything else to shame. The chipset is quickly upstaged by its older brother the Ti 4600 that comes quite close to that magical score of 300 fps (290.4 fps).

Integrated Graphics

Graphics chipsets integrated onto motherboards have always come as an ideal solution for the budget-conscious consumer. When Intel came out with a motherboard based on the 810 chipset, they became quite popular in the value category. The main reason for this was that the boards were priced low and the graphics performance offered by them enabled the user to play most of the games available then. The only drawback was that your upgrade choices were limited by the lack of an AGP slot on these motherboards. Intel followed up the success of the 810 chipset with another successful entry—the 815 chipset. This chipset offered better performance



than the 810 and boards based on these chipsets even sported an AGP slot for future upgrades. The latest offering from Intel, the 845GL chipset, aims to further

increase performance levels in the onboard graphics arena.

Both the current chipsets, the 815 and the 845GL, performed adequately through our speed tests—in *Quake III Arena*, you can expect decent performance with the 845GL at lower resolutions as reflected by its score of 105 fps. Also, even though its 3DMark2001 score bordered on the lower end of the spectrum, the 845GL with 1345 3DMarks can hold its own against GeForce2 MX200 based solutions. Not too bad considering that this onboard video chip misses out on the more whiz-bang features that are available in the current generation 3D chipsets.